

HAT-P-32b

R-0051

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_171107 · created 2026-07-09T08:05:01+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458064.7095 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.015
Transit depth (Rp/Rs)^2	2.04 +/- 0.42 [%]
Orbital Inclination (inc)	89.67 +/- 1.49
Airmass coefficient 1 (a1)	0.931 +/- 0.017
Airmass coefficient 2 (a2)	0.058 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.78 %
Best Comparison Star	#1 - [584, 75]
Optimal Aperture	4.19
Optimal Annulus	10.98
Transit Duration (day)	0.1291 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.015 vs 0.1489 (deviation 0.39σ)
Duration fit vs published	0.1291 d vs 0.1304 d (deviation 0.69σ)
Mid-time O–C	-2.0 min
Depth SNR	9.5
Residual scatter	1.78 %

Light curve

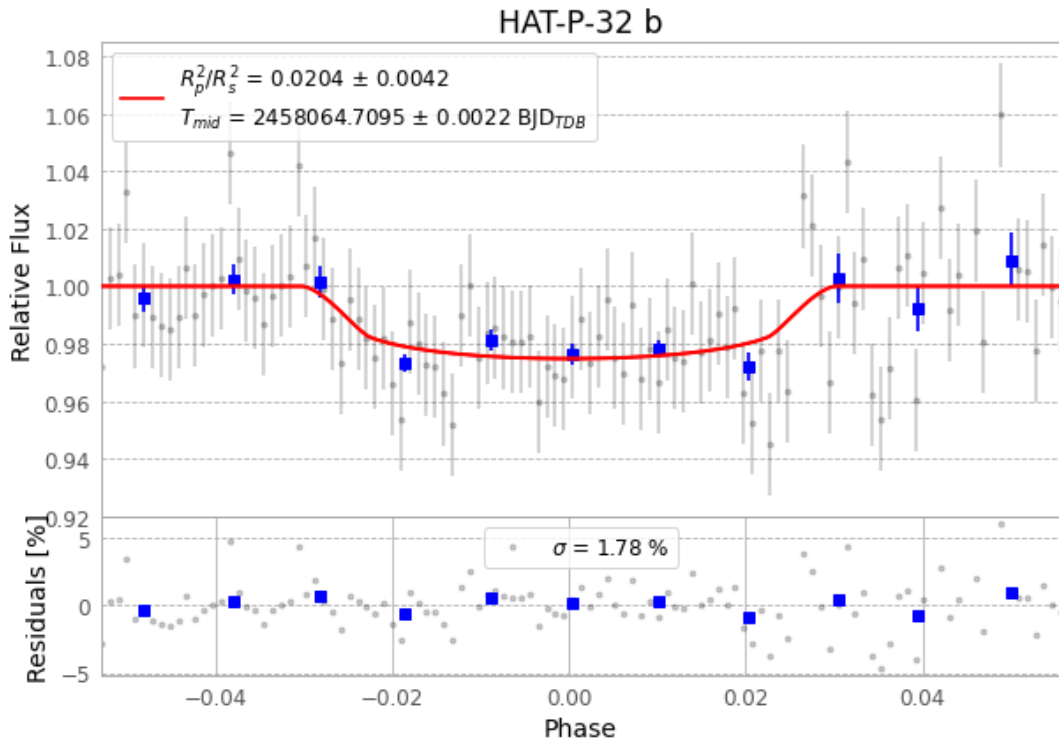


Figure 1 — FinalLightCurve_HAT-P-32 b_2017-11-06.png (SHA-256 c02557d9b54df4da...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [381, 234], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7628534d7c522d52...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

113 FITS frames (74 MB), HATP-32171107020913.FITS → HATP-32171107074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-171107.log (1a3945125665...), FinalLightCurve_HAT-P-32 b_2017-11-06.png (c02557d9b54d...), FinalLightCurve_HAT-P-32 b_2017-11-06.csv (b5662bd6c648...)

Compute resources

Wall time 88.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.